

Project Overview – US Visa Prediction Using Python

This project focuses on building a **US Visa Prediction System using Python and Machine Learning** to predict whether a visa application will be **Certified (Approved)** or **Denied** based on applicant and employer information.

The project includes:

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Feature engineering
- Model training and evaluation
- Prediction and business insights

The goal is to help identify factors that influence visa approval and support data-driven decision making.

Dataset Summary

Dataset: EasyVisa Dataset

The dataset contains historical US visa application records.

Main Columns:

- case_id → Unique application ID
- continent → Applicant continent
- education_of_employee → Education level
- has_job_experience → Work experience
- requires_job_training → Training requirement
- no_of_employees → Company employee count
- yr_of_estab → Company establishment year
- region_of_employment → Employment region
- prevailing_wage → Salary offered
- unit_of_wage → Wage type
- full_time_position → Full-time job indicator
- case_status → Target variable (Certified / Denied)

Exploratory Data Analysis (EDA) Using Python

Data Preprocessing

- Checked missing values
- Removed duplicate records
- Converted categorical columns
- Encoded target variable
- Standardized dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   case_id               25480 non-null  object
 1   continent             25480 non-null  object
 2   education_of_employee 25480 non-null  object
 3   has_job_experience     25480 non-null  object
 4   requires_job_training 25480 non-null  object
 5   no_of_employees       25480 non-null  int64
 6   yr_of_estab           25480 non-null  int64
 7   region_of_employment  25480 non-null  object
 8   prevailing_wage       25480 non-null  float64
 9   unit_of_wage          25480 non-null  object
10   full_time_position    25480 non-null  object
11   case_status           25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

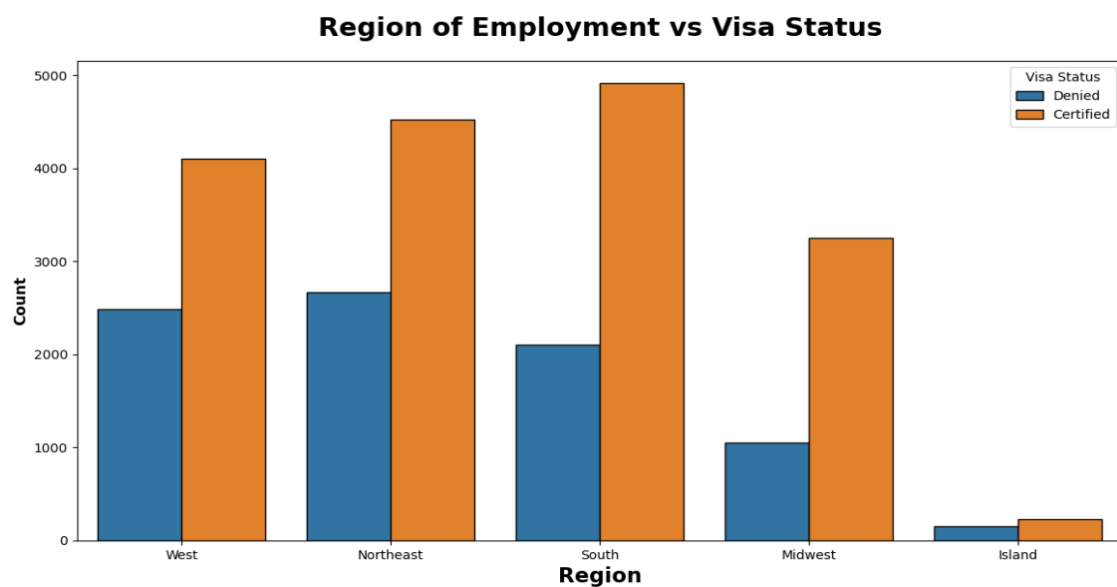
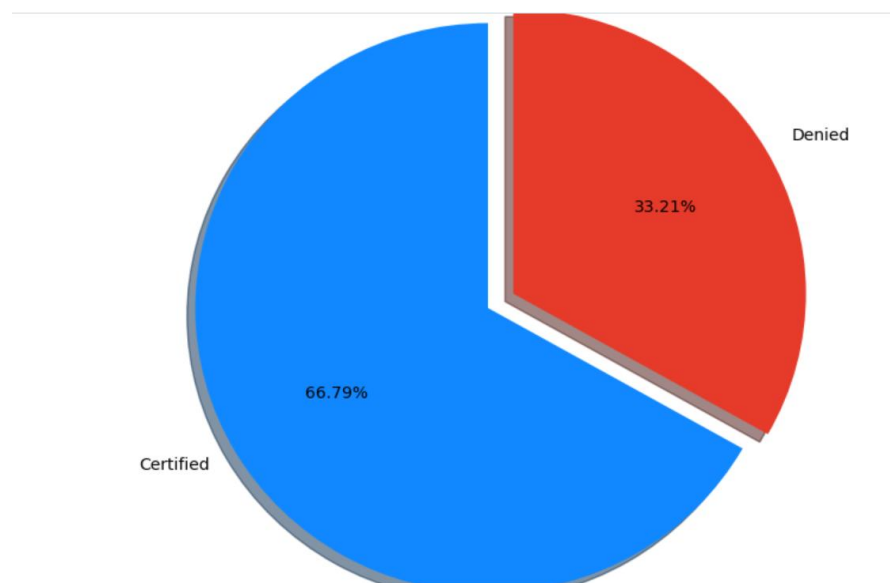
```
We have 3 numerical features : ['no_of_employees', 'yr_of_estab', 'prevailing_wage']
```

```
We have 9 categorical features : ['case_id', 'continent', 'education_of_employee', 'has_job_experience', 'requires_job_training', 'region_of_employment', 'unit_of_wage', 'full_time_position', 'case_status']
```

Analysis Performed

- Visa approval distribution
- Education vs visa approval
- Experience vs certification
- Salary impact analysis
- Region-wise approval trends

- Company size analysis



Key EDA Insights

- Applicants with **higher education** received more certifications.
- Candidates with **job experience** had better approval rates.
- **Full-time positions** showed higher certification probability.
- Higher **prevailing wages** increased approval chances.
- Employment region influenced certification outcomes.

Business Recommendations

1. Hire Skilled Candidates

Prefer applicants with relevant education and work experience.

2. Offer Competitive Salaries

Higher prevailing wages may improve visa approval chances.

3. Encourage Full-Time Roles

Full-time applications tend to perform better.

4. Improve Data Quality

Ensure complete and accurate application details.

5. Use Machine Learning Before Submission

Predict approval probability before filing applications.

6. Monitor Regional Trends

Focus hiring strategies based on region-wise approval performance.

Outcome

This project helps organizations **reduce risk, improve hiring decisions, and increase visa approval success** using **Python and Machine Learning**.